

UGUT, Russian Federation

WMO: 239460

Lat: 60.500N

Lon: 74.017E

Elev: 48

StdP: 100.75

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.6	(c) -36.2	(d) -42.2	(e) 0.1	(f) -39.0	(g) -39.1	(h) 0.1	(i) -35.8	(j) 6.2	(k) -9.5	(l) 5.7	(m) -10.5	(n) 0.9	(o) 240	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.3	18.7	27.4	18.2	25.5	17.5	20.7	25.7	19.7	24.6	18.8	23.2	2.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	22.2	18.1	13.1	21.5	17.0	12.2	20.9	60.2	25.9	56.6	24.8	53.7	23.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.4	5.7	5.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-43.0	22.4	1.6	1.7	-44.2	23.6	-45.1	24.6	-46.0	25.6	-47.1	26.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.7	-20.1	-17.1	-7.9	0.5	6.7	16.0	18.2	14.7	7.9	-0.6	-11.9	-16.2		
	DBStd	14.73	9.25	9.22	7.01	5.60	5.57	5.24	3.81	3.92	4.26	5.27	8.50	9.13		
	HDD10.0	4582	934	759	556	287	137	12	0	7	91	329	658	813		
	HDD18.3	7054	1192	992	814	534	364	103	51	124	314	586	908	1072		
	CDD10.0	664	0	0	0	2	33	193	253	153	28	2	0	0		
	CDD18.3	94	0	0	0	0	2	35	46	11	1	0	0	0		
	CDH23.3	929	0	0	0	0	43	362	434	89	1	0	0	0		
	CDH26.7	237	0	0	0	0	8	99	118	12	0	0	0	0		
	WSAvg	2.3	2.0	2.0	2.5	2.7	2.8	2.3	2.3	2.0	2.2	2.4	2.3	2.0		
(15) Precipitation	PrecAvg	603	30	25	33	32	56	70	66	82	64	51	48	39		
	PrecMax	809	67	46	94	65	163	160	137	171	120	106	74	91		
	PrecMin	458	6	11	1	1	21	3	7	21	13	19	15	12		
	PrecStd	99	15	9	19	14	31	29	33	38	31	23	16	17		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	2.3	6.3	17.7	27.7	31.5	32.3	28.3	22.4	15.8	3.1	0.0		
		MCWB	-1.5	0.6	1.2	11.6	15.5	19.6	21.3	18.9	14.4	9.4	2.5	-0.5		
	2%	DB	-3.3	0.1	3.7	13.4	23.6	29.4	29.6	25.6	19.0	10.8	1.5	-1.2		
		MCWB	-3.9	-0.7	0.4	8.2	14.4	19.4	18.2	17.8	13.0	8.0	0.9	-1.7		
	5%	DB	-5.4	-2.2	2.2	10.8	20.3	27.4	27.8	23.9	16.4	8.3	0.2	-2.8		
		MCWB	-6.0	-3.0	-0.2	6.2	13.2	18.7	18.1	17.3	12.0	6.9	-0.3	-3.4		
	10%	DB	-7.4	-4.9	1.0	8.8	16.1	25.3	25.8	22.2	14.3	6.2	-0.8	-4.9		
		MCWB	-7.9	-5.8	-0.9	4.8	11.0	17.7	17.6	16.8	11.3	4.7	-1.4	-5.4		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	0.8	2.3	11.8	17.5	23.1	22.4	20.6	15.6	10.1	2.7	-0.4		
		MCDB	-1.4	1.7	4.7	16.7	25.7	27.8	30.1	24.9	19.5	13.5	2.9	-0.1		
	2%	WB	-3.9	-0.7	1.2	9.0	15.2	21.0	20.7	19.5	14.0	8.2	1.0	-1.7		
		MCDB	-3.3	0.0	3.1	12.5	21.3	26.2	25.4	23.4	17.1	10.4	1.4	-1.2		
	5%	WB	-6.0	-2.8	0.2	6.8	13.4	19.9	19.6	18.5	12.8	6.8	-0.1	-3.3		
		MCDB	-5.5	-2.1	1.9	10.4	19.6	25.9	24.1	22.3	15.7	8.6	0.1	-2.9		
	10%	WB	-8.0	-5.8	-0.9	5.0	11.5	18.8	18.8	17.3	11.5	4.9	-1.3	-5.3		
		MCDB	-7.5	-4.9	0.8	7.9	16.1	23.6	23.1	20.8	14.3	6.0	-0.8	-4.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	9.1	9.6	9.5	9.7	10.2	10.2	9.4	7.3	5.2	7.3	6.8		
		MCDBR	7.8	8.2	8.3	12.3	15.7	13.6	13.9	12.6	10.5	7.4	6.1	6.5		
		MCWBR	7.6	7.9	6.3	7.9	8.2	5.9	4.8	6.0	5.5	5.3	6.1	6.4		
	5% WB	MCDBR	7.8	8.1	7.4	11.8	14.6	12.2	10.8	10.4	9.3	7.2	6.4	6.0		
		MCWBR	7.6	7.8	6.0	8.1	8.7	6.3	4.6	5.9	5.6	5.5	6.4	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.306	0.359	0.379	0.392	0.403	0.425	0.413	0.362	0.331	0.278	0.250		
	taud		1.979	2.008	1.947	2.065	2.277	2.327	2.259	2.266	2.370	2.261	2.052	1.923		
	Ebn at Noon		446	645	728	804	830	826	794	769	751	631	425	292		
	Edh at Noon		47	84	125	134	118	114	120	110	83	66	43	32		
(44) All-Sky Solar Radiation	RadAvg		0.44	1.34	2.81	4.33	4.80	5.30	5.05	3.59	2.23	0.98	0.42	0.22		
	RadStd		0.04	0.11	0.21	0.42	0.42	0.49	0.56	0.35	0.26	0.11	0.09	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-368	N/A	N/A

Nomenclature: See separate page